

COAST: Supplementary Material

Anonymous WACV submission

Paper ID ****

001	1. Complete model results	References	038
002	The main paper focuses on six regression and spatial	[1] Taejin Jeong, Joohyeok Kim, Jinyeong Kim, Chanyoung	039
003	backbones for which global and local FiLM form a di-	Kim, and Seong Jae Hwang. Feast: Fully connected ex-	040
004	rectly matched intervention. For transparency, this ap-	pressive attention for spatial transcriptomics. In IEEE/CVF	041
005	pendix reports every evaluated architecture, including	Conference on Computer Vision and Pattern Recognition	042
006	the neutral TRIPLEX result and the negative BLEEP	(CVPR), 2026. 1	043
007	result. No model was removed from the experimental	[2] Yaxuan Song, Jianan Fan, Hang Chang, and Weidong	044
008	record.	Cai. Gene-dml: Dual-pathway multi-level discrimination for	045
		gene expression prediction from histopathology images. In	046
		IEEE/CVF Winter Conference on Applications of Computer	047
		Vision (WACV), 2026. 1	048
009	Paired tests. Wilcoxon tests on paired fold–seed val-		
010	ues for the best LOOO variant give $p < 10^{-4}$ for		
011	ST-Net and DeepSpaCE, $p = 0.0002$ for HisToGene,		
012	$p = 0.074$ for Hist2ST, $p = 0.032$ for the MLP probe,		
013	and $p = 0.390$ for TRIPLEX. BLEEP’s global inter-		
014	vention decreases performance ($p = 0.016$). These		
015	preliminary values must be confirmed using validation-		
016	selected checkpoints before submission.		
017	Recent-baseline execution status. We additionally in-		
018	tegrated the official Gene-DML [2] and FEAST [1] code		
019	releases. Because COAST fixes UNI embeddings and		
020	fold-specific 50-gene panels, the reported Gene-DML-		
021	style row retains its spot, neighborhood, and global		
022	streams with transformer fusion and auxiliary stream-		
023	wise regression, but does not claim an end-to-end re-		
024	production of the original system. FEAST results are		
025	not reported: dense attention exceeded GPU memory		
026	for several large slides, and other runs produced unde-		
027	efined validation correlations. Omitting those numbers		
028	avoids converting an engineering failure into an appar-		
029	ent scientific comparison. Memory-safe FEAST reruns		
030	remain required for the final version.		
031	2. Complete unconditioned benchmark		
032	The final appendix will additionally contain the full		
033	HEST and STImage-1K4M model tables, per-organ		
034	scores, per-seed results, alternative metrics, confidence		
035	intervals, and threshold-sensitivity curves. This sepa-		
036	ration keeps the main paper focused while preserving		
037	all comparisons needed for audit.		

Backbone	LOOO				POOLED			
	None	Global	Local	Best Δ	None	Global	Local	Best Δ
ST-Net	.3588	.3493	.3969	+.0380	.4169	.3286	.5886	+.1717
DeepSpaCE	.4521	.4590	.4672	+.0151	.7834	.7890	.7908	+.0074
HisToGene	.4384	.4526	.4522	+.0142	.7775	.7830	.7881	+.0106
Hist2ST	.4126	.4147	.4183	+.0057	.7659	.7697	.7702	+.0043
MLP probe	.4530	.4589	.4588	+.0059	.7871	.7878	.7930	+.0059
Gene-DML-style [†]	.4527	.4597	.4576	+.0070	.7856	.7892	.7852	+.0036
TRIPLEX	.4639	.4610	.4619	-.0020	.7909	.7922	.7932	+.0023
BLEEP	.4255	.4161	–	-.0093	.7722	.7696	–	-.0027

Table 1. Complete HEST cross-architecture conditioning screening (three-seed means). Bold marks the best variant within each backbone and regime. BLEEP has no compatible local intervention. [†]Feature-matched COAST adaptation of Gene-DML; LOOO None and POOLED Local contain two seeds, and the remaining Gene-DML cells contain three.